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REPRODUCTIVE HEALTH

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADE Adverse Drug Event

BID Twice Daily

COC Combined Oral Contraceptive

CNS Central Nervous System

FSH Follicle-Stimulating Hormone

GnRH Gonadotropin-Releasing Hormone

hCG Human Chorionic Gonadotropin

HG Hyperemesis Gravidarum

IU International Units

IUD Intrauterine Device

IUI Intrauterine Semination

IUS Intrauterine System

IV Intravenous

IVF In-Vitro Fertilization

LH Luteinizing hormone

NRT Nicotine Replacement Therapy

NTD Neural Tube Defects

N/V Nausea/Vomiting

NVP Nausea/Vomiting in Pregnancy

OPK Ovulation Predictor Kits

PCOS Polycystic Ovary Syndrome

PEG Polyethylene Glycol

PO By Mouth

PPD Postpartum Depression

PRN As needed

SC Subcutaneous

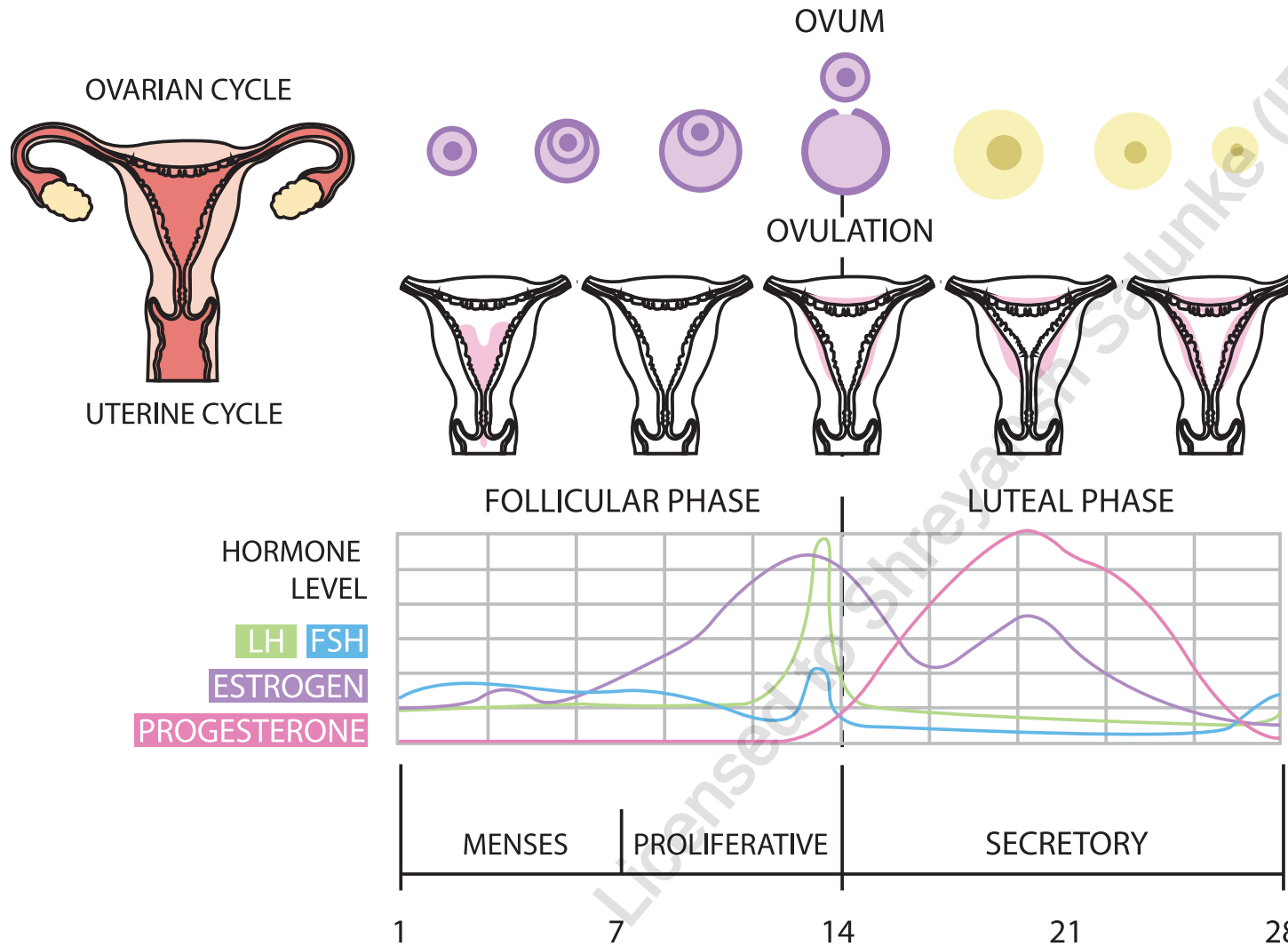
PREGNANCY AND FERTILITY

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FEMALE REPRODUCTIVE CYCLE

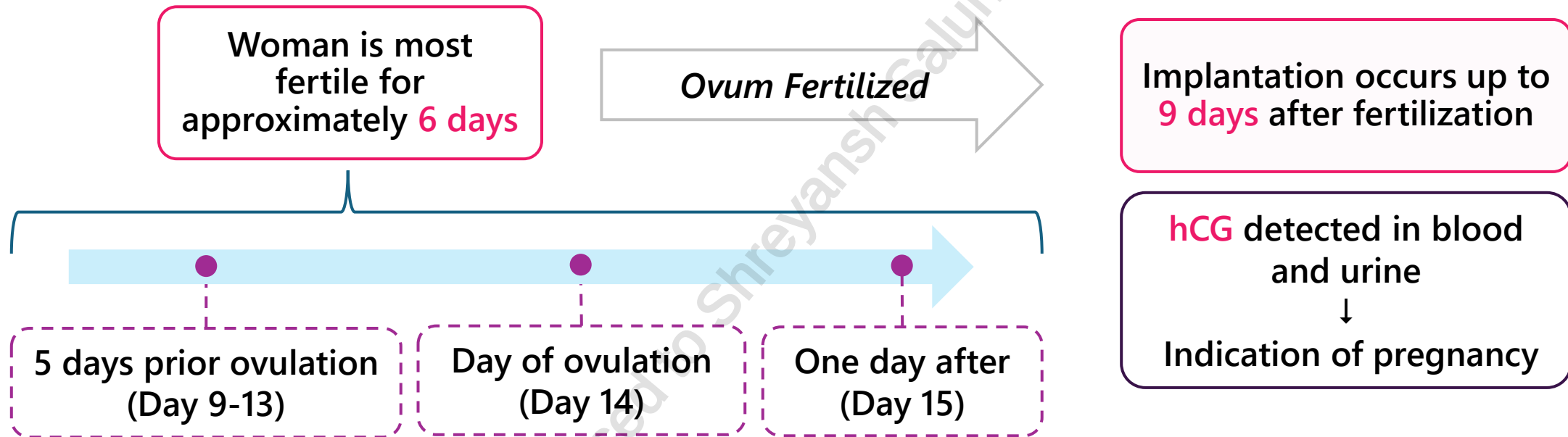


Ovulation tests vs. Pregnancy tests

- Ovulation predictor kits (OPK) detect **luteinizing hormone (LH)** surge
- Pregnancy tests detect **human chorionic gonadotropin (hCG)**

FERTILITY

- Female: once ovulation occurs, egg is viable for **24 hours**
- Male: upon ejaculation, sperm is viable for up to **5 days**



FERTILITY TESTING

Popular methods:

- Ovulation predictor kits (OPK)
- Basal body thermometers
- Calendar method
- Monitor cervical mucus

Other methods:

- Endometrial biopsy
- Midluteal serum progesterone level
- Saliva microscopic exam

Basal Body Temperature

- The **lowest** body temperature reached at **complete rest**
- Measured immediately after waking and before getting up

INFERTILITY

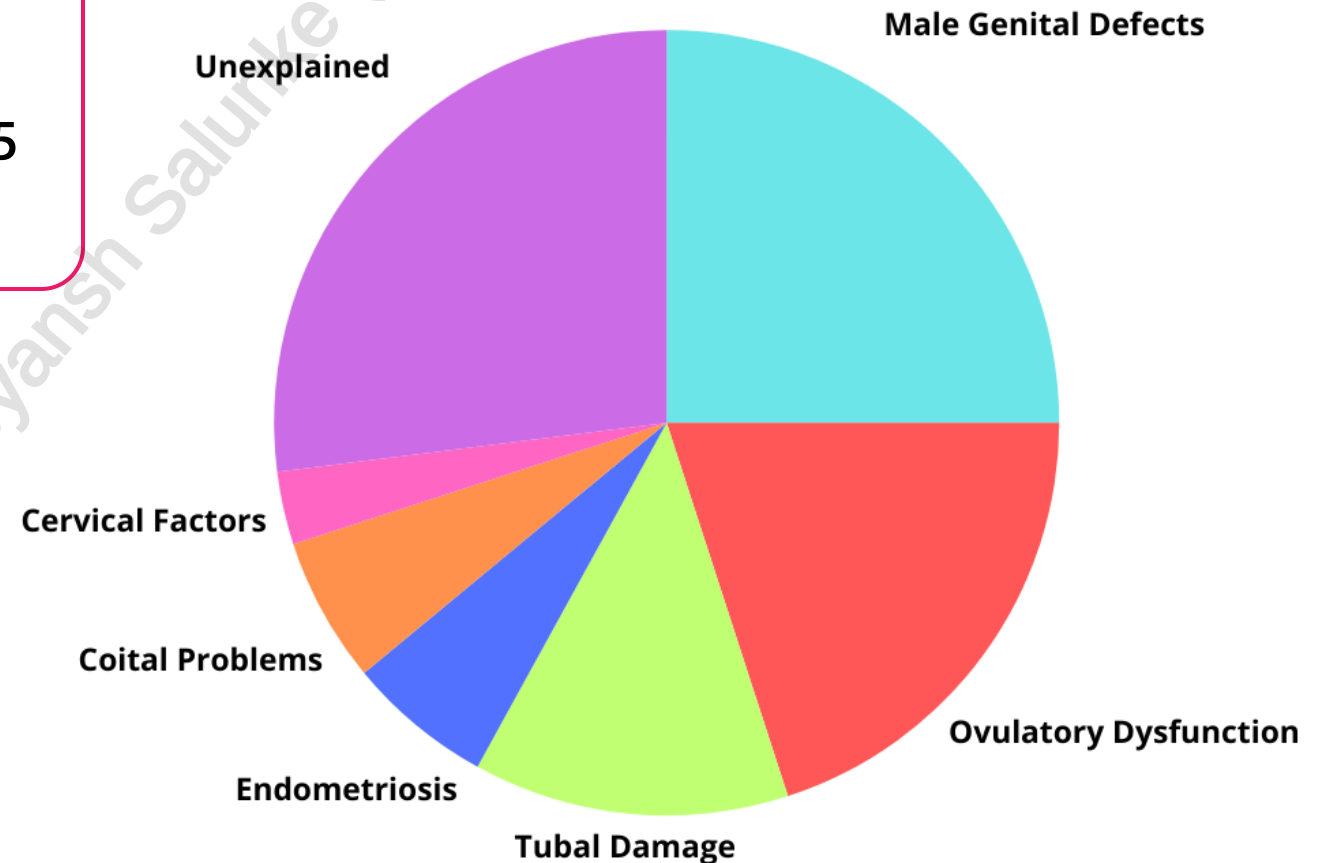
Infertility

- The inability to conceive with regular unprotected intercourse
- Diagnosed after 12 months in women <35
OR after 6 months in women ≥35 years

Diagnostics include:

- Physical history and exam
- Menstrual history
- Lab tests
- Laparoscopy, ultrasonography, sonohysterography, hysterosalpingography
- Semen analysis

Causes of Infertility



NON-PHARMACOLOGICAL MEASURES: INFERTILITY

Treat reversible etiologies:

- Smoking cessation
- Reduce alcohol/caffeine intake
- Review drug therapy
- Review timing of coitus
- Weight modulation if overweight/underweight (females)

Attempt to overcome irreversible ones:

- Consider intrauterine insemination (IUI) and/or in-vitro fertilization (IVF)

PHARMACOLOGICAL THERAPY: INFERTILITY

Female Infertility

- Ovarian induction
 - Aromatase inhibitors (e.g. letrozole)
 - Gonadotropin therapy (e.g. LH, FSH)
- Metformin for hyperinsulinemia in PCOS
- Dopamine agonists for hyperprolactinemia (e.g. bromocriptine, cabergoline)
- Surgical and drug therapy for structural /chronic disorders (e.g. tubal/endometriosis)

Male Infertility

- Gonadotropin therapy (e.g. LH, FSH) to enhance spermatogenesis and treat hypogonadism
- Dopamine agonists for hyperprolactinemia (e.g. bromocriptine, cabergoline)
- Surgical interventions for structural and functional disorders of the reproductive system

PHARMACOLOGICAL THERAPY: INFERTILITY

- **Letrozole PO 2.5 mg daily (days 3-7 of cycle)**
 - Aromatase inhibitor blocks conversion of androgen to estrogen = increased FSH = ovarian follicle growth
 - **ADEs: hot flashes, joint pain, nausea, fatigue**
- **Gonadotropins**
 - LH and FSH → ovarian follicle growth
 - Example: follitropin alfa (Gonal-F®)
 - **ADEs: headache, ovarian hyperstimulation/cysts, higher risk of multiple pregnancy**
- **Human Chorionic Gonadotropin (hCG)**
 - Added midcycle to induce ovulation
 - Example: choriogonadotropin alfa (Ovidrel®) SC injection, chorionic gonadotropin (Pregnyl®) IM injection
 - **ADEs: ovarian hyperstimulation/cysts, higher risk of multiple pregnancy**

PREGNANCY TESTING

Indication of pregnancy: **hCG** is detected in blood and urine

Common errors leading to false results:

Test kit	Timeframe	Contamination	Other
• Expired test kit • Not following specific manufacturer instructions	• Too early after conception when hCG levels are too low to be detected • Too late (after 60–70 days when hCG levels decline) • Too soon after an abortion	• Soap, blood or protein in the urine sample • Cloudy, pink or red urine • Strong urine odour • Warm or hot water rinses the test surface before or after sampling	• Use of hormones, corticosteroids, and fertility drugs containing hCG • Note: letrozole will not affect pregnancy tests

CLINICAL PEARLS: PREGNANCY TESTING

- Test **after** the expected period date
 - 3rd generation pregnancy tests (home urine kits) can detect hCG as early as 5-6 days before a missed period, but false negatives are more common
 - Testing 4 days before the expected period can detect a pregnancy in only **51% of pregnant women**
 - It is recommended to test at least **1 day after** a missed period
- **First-morning urine** has highest hCG concentration
- Collect **midstream** urine sample to avoid contamination
- Follow specific manufacturer instructions

PRENATAL AND POSTPARTUM CARE

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PRENATAL NUTRITION

General

- Start **prenatal multivitamin with vitamin B12** at least 2-3 months before conception, continue throughout pregnancy and for 4-6 weeks postpartum or as long as breastfeeding

Iodine: 220-250 mcg

- Supports fetal neurologic development & maternal metabolism
- Diet: milk, eggs, cheese
- Supplement with 150 mcg/day
- Avoid chronic use of >500-1100 mcg/day due to risk of fetal hypothyroidism

Iron: 27 mg

- Supports fetal brain development & reduces risk of maternal **anemia**
- Diet: meat, poultry, tofu, lentils
- Supplement with 16-20 mg/day with vitamin C
- Can exacerbate N/V, constipation

Calcium: 1000 mg

- Supports fetal bone growth
- Diet: 2 dairy servings/day
- **OR if <2 servings/day supplement with 1000 mg/day, with 600 IU Vitamin D**
- Can exacerbate constipation

PRENATAL NUTRITION- FOLIC ACID

- Essential for CNS development→ reduces risk of neural tube defects (NTDs)
- Duration: supplement with folic acid at least **3 months preconception, continue throughout pregnancy, and for 4-6 weeks postpartum or if breastfeeding continues**

Low risk factors	No family history	0.4 mg daily as directed
Moderate risk of NTDs	• Personal or family history of folate-sensitive congenital anomalies (e.g. cleft palate, congenital hydrocephaly) • Non-immediate family history of NTD • Maternal diabetes, dialysis, alcohol use disorder, GI malabsorption (e.g. Crohn's/celiac disease, gastric bypass), folate-inhibiting medication use*	1 mg daily preconception until 12-week gestational age (2 nd trimester) 0.4 mg daily as directed thereafter
High risk of NTDs	• Previous NTD pregnancy • First-degree relative with NTD • Personal history of NTD	4 – 5 mg daily preconception until 12-week gestational age 0.4 mg daily as directed thereafter

***Folate-inhibiting medications:** carbamazepine, cholestyramine, metformin, methotrexate, phenobarbital, phenytoin, primidone, sulfasalazine, triamterene, trimethoprim, valproic acid

LIFESTYLE CHANGES IN PREGNANCY

Exercise

- Moderate exercise appears to be safe
- Avoid contact sports and intense activity, especially in the first trimester

Diet

- Avoid eating raw and undercooked meat, eggs, seafood and unpasteurized dairy
- Limit caffeine ≤ 300 mg/day

Alcohol

- **Avoid alcohol use.** Teratogenic agent with risks throughout pregnancy
- Stillbirth and fetal alcohol spectrum disorder risks increase

Cannabis

- **Avoid cannabis use.** Limited data suggests possible harm
- Prenatal exposure associated with neurocognitive and behavioural deficits

Tobacco

- **Avoid tobacco use.** Preterm deliveries/deaths, growth restriction, and sudden infant death syndrome linked to cigarette use. **Offer smoking cessation (NRT)**

NAUSEA AND VOMITING OF PREGNANCY (NVP)

- Referred to as “**morning sickness**”
- Occurs in approximately **50-80% of pregnancies**
 - Typically occurs during **first trimester** and generally subsides by 16 weeks, but NVP can occur at any time and may continue throughout pregnancy
- Pathophysiology is unclear
- Severe, persistent NVP is called **hyperemesis gravidarum (HG)**
 - Occurs in approximately 2% of pregnancies
 - Leads to weight loss, electrolyte imbalances, dehydration, nutritional deficiency

NON-PHARMACOLOGICAL MEASURES: NVP

- Eat whatever tolerated, as long as pregnancy-safe
- **Avoid triggers** such as strong food odours or spicy, fried, and greasy foods
- Take supplements **after meals** or at **bedtime**
- Substitute iron-containing prenatal vitamin with folic acid (1st trimester)
- Prevent dehydration and electrolyte imbalance
 - Adequate fluid intake
 - **Rehydration solutions** (if severe and/or accompanied by diarrhea)
- P6 acupressure appears safe and may help

PHARMACOLOGICAL MEASURES- NVP

Mild

- **1st line:** **Pyridoxine** (Vitamin B6) 10 mg 4x/day monotherapy or **Diclectin®** (pyridoxine/doxylamine 10 mg/10 mg 4x/day)
- Add **Gravol®** (dimenhydrinate) 50 mg Q4-6H PRN

Moderate /
Persistent

- Add any of the following: (may require IV fluids prior, if dehydrated)
- **Metoclopramide** 5-10 mg Q8H
 - Phenothiazines (**chlorpromazine**, **prochlorperazine** or **promethazine**)

Severe /
Refractory

- Add: Avoid during 1st trimester
- **Ondansetron** 8 mg Q12H or 4 mg Q8H PRN
 - Refractory: **methylprednisolone IV**

May add at any time:

- **Pyridoxine**
25 mg Q8H
- **Ginger**
250 mg Q6H

POSTPARTUM CARE

Postpartum Perineal Care and Post-Episiotomy Pain

- **Non-Pharmacological:** localized cooling, perineal squirt bottles, Kegel exercises
- **Pharmacological:** non-opioid analgesics are 1st line

Postpartum Blues and Depression

- Up to 85% of mothers experience “baby blues”
- If severe enough to be classified as **postpartum depression (PPD)**, treat as major depressive disorder with antidepressants to prevent developmental issues
- Psychotherapy is effective — assess benefits vs. cost

POSTPARTUM CONTRACEPTIVE USE

Lactational amenorrhea (breastfeeding that delays ovulation)

- Effective birth control method if patient is **<6 months postpartum**, **breastfeeding >75%** of all feedings, and remains **amenorrheic**
- Difficult to maintain, not as reliable

Can use immediately post-partum, regardless of breastfeeding status:

- **Progestin-only methods**
- **IUD/IUS**

Avoid **combined estrogen-progestin oral contraceptives (COC)**:

- If not breastfeeding: until **3-4 weeks post-partum**
- If breastfeeding: until breastfeeding established, usually after **6 weeks post-partum**

Emergency Contraception

- Plan B® (levonorgestrel 1.5 mg) can be used within 72 hours
- Ella® (ulipristal acetate 30mg) can be used within 5 days
 - Breastfeeding not recommended for 1 week after (breast milk should be pumped and discarded)

CASE

EM is a 30-year-old woman with regular menstrual cycles and has been trying to conceive for the past year without success. After discussing her situation with her pharmacist, EM is advised to use an over-the-counter ovulation test. EM is unsure about the best time during her cycle to begin testing.

EM has a medical history of diabetes. Her pharmacy profile shows she takes metformin 500 mg BID. Upon inquiry, she is also taking a multivitamin, calcium 500 mg once daily, vitamin D 2000 IU once daily, polyethylene glycol (PEG) 17 g once daily as needed, and cetirizine 10 mg once daily.

What is the best recommendation for when EM should start using the ovulation predictor kit (OPK)?

- a) Day 5
- b) Day 10
- c) Day 14
- d) Day 15



CASE

EM successfully conceives and is now 6 weeks pregnant. She is visiting her local pharmacy seeking advice on prenatal vitamins and supplements to support a healthy pregnancy. Her aunt and cousin have a history of neural tube defects (NTDs) and EM is concerned about ensuring she gets the right nutrients.

EM has a medical history of diabetes. Her pharmacy profile shows she takes metformin 500 mg BID. Upon inquiry, she is also taking a multivitamin, calcium 500 mg once daily, vitamin D 2000 IU once daily, polyethylene glycol (PEG) 17 g once daily as needed, and cetirizine 10 mg once daily.

Which of the following is the most appropriate recommendation for EM regarding folic acid supplementation at this time?

- a) 0.4 mg daily
- b) 0.6 mg daily
- c) 1 mg daily
- d) 4 mg daily



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CHANGE LOG

April 2025

- Added legend
- Slide 12: Clarified contamination section
- Slide 15: Added daily requirements, food sources, iodine section, updated calcium information, removed folic acid
- Slide 16: Added indication for folic acid
- Slide 20: 'Avoid during 1st trimester' for both ondansetron and methylprednisolone
- Added references

May 2025

- Slide 6: Updated fertility window from 7 days to 'approximately 6 days'
- Slide 16: Updated folic acid recommendations
- Slide 25: Updated references

August 2025

- Content reviewed, no edits made

November 2025

- Minor formatting edits; content changes made

March 2026

- Slide 2: Updated NAPRA competencies slide
- Slide 3: Updated legend
- Slide 11: Added additional IM example of hCG product; changed title of 'GnRH agonists' to 'gonadotropins'
- Slide 16: Slight edits based on SOGC guidance
- Slide 20: Minor edits
- Minor formatting edits throughout